

MUHAMMAD MAROOF

Associate Software Engineer | Full Stack Developer | AI Engineer

Lahore, Pakistan | +92 305 1526463 | Email: mmaroof341@gmail.com

LinkedIn: linkedin.com/in/mmaroof11/ | GitHub: github.com/muhammadmaroof11

PROFESSIONAL SUMMARY

Performance-driven Full Stack AI Developer specializing in full-stack applications, large language models (LLMs), and agentic workflow automation with great attention to details. Robust experience building scalable software applications, designing multi-agent systems (MASS), implementing Model Context Protocol (MCP), and managing the software development lifecycle (SDLC) under Agile methodologies. Proficient in Python, JavaScript, and Node.js backend environments. Proven capability in prompt engineering, technical evaluation of local vs. cloud LLMs, and integrating Retrieval-Augmented Generation (RAG) frameworks to optimize contextual data delivery.

TECHNICAL SKILLS

- **AI, LLMs & Automation:** Agentic AI, Gen AI, OpenAI Integration, Multi-Agent Systems (MASS), LLM APIs (OpenAI, Anthropic), Model Context Protocol (MCP), Prompt Engineering & Evaluation, n8n Automation, Workflow Orchestration, Roo Code, Claude Code.
- **RAG & Core Concepts:** Retrieval-Augmented Generation (RAG), Semantic Search Concepts, Knowledge Bases, Embeddings & Vector Databases (Familiarity with Pinecone, pgvector).
- **Programming Languages:** Python, JavaScript (ES6+), C++, Dart, Java, SQL, HTML5, CSS3.
- **Frameworks & Libraries:** Node.js, Express.js, Flask, React.js, Vue 3, Flutter, Ionic Framework, Tailwind CSS, Bootstrap.
- **Databases:** MongoDB (NoSQL), SQL, Supabase, Firebase.
- **Cloud, DevOps & Tools:** Git, Version Control, Docker, Linux, CI/CD, Cloud Platforms (AWS, GCP, Azure), CMake.
- **Core Competencies:** Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful APIs, clean code, software engineering best practices, debugging, code reviews.

WORK EXPERIENCE

Byte Force IT Solutions | Lahore, Pakistan *Full Stack Developer* | Feb 2026 – Present

- Design, develop, test, and maintain scalable software applications, integrating RESTful APIs and backend services using modern technologies.
- Engineer and deploy autonomous workflow pipelines utilizing n8n to connect APIs with outside services, managing containerized systems.
- Participate in code reviews, sprint planning, and Agile ceremonies, writing clean, reliable code and investigating system bugs.

Sairay.pk | Remote *Full Stack Developer (Freelance)* | Nov 2025 – Feb 2026

- Architected a robust, scalable REST API utilizing Python (Flask) to handle dynamic enterprise product inventories.
- Accelerated frontend product discovery workflows by 40% through optimized React.js state-filtering algorithms.

Privacy-First AI Matrimonial Platform (Freelance Work) *MERN Stack (MongoDB, Express.js, React.js, Node.js), Three.js, Google Cloud Vision API, Meta WhatsApp Business API, FCM, SMTP*

- Architected a scalable bilingual (Urdu/English) application using MongoDB, integrating Google Cloud Vision API for CNIC OCR, SMTP-based Email OTP authentication, Firebase Cloud Messaging (FCM), and the Meta WhatsApp Business API to deliver secure identity verification, real-time notifications, and localized profile matching.

PROJECTS

Bytely Personal Assistant (SaaS Platform) *Vue 3, Ionic Framework, Pinia, Firebase, Capacitor, Node.js*

- Architected a unified AI chat orchestrator for 7 specialized AI agents, reducing component bloat by 50% through composable logic extraction while integrating REST APIs, dynamic prompt management, real-time rendering (DOMPurify, marked), and an offline-first AES-256 encrypted IndexedDB cache with background sync.

Autonomous AI Agent Workflows & Model Evaluation *n8n, Python, Local/Cloud LLMs, MCP*

- Designed multi-agent systems (MASS) and custom automation pipelines using n8n to orchestrate complex backend workflows.
- Configured Model Context Protocol (MCP) servers linking developer tools (Roo Code, Continue) with LLMs for autonomous codebase context injection.
- Conducted latency and output benchmarks comparing local offline LLMs against cloud APIs (OpenAI/Anthropic) to optimize deterministic prompt engineering.

Prompt Prettier (Open-Source Contribution) *Node.js, Agentic AI, Local LLMs*

- Architected a local VS Code extension to structure, compile, and optimize developer prompts for AI agents.
- Implemented Gen AI workflows requiring rigorous application of Data Structures & Algorithms and clean code execution.

Ezi Services (Enterprise Optimization) *Full Stack Web Production*

- Single-handedly re-architected an enterprise high-performance web application within a 1-week Agile timeline.
- Executed deep core-performance tuning, driving the platform's Google Lighthouse performance score to 98% and reducing page load times by 55%.

ADHD-Focused Desktop Task Manager *C++, CMake*

- Engineered a high-performance, native desktop application using C++ and CMake, featuring complex synchronization mechanics and modular task categorization.

EDUCATION

Bachelor's Degree in Computer Science University of Education, Lahore, Pakistan | 2023 – Present

- Currently in Final Year (Post-ADP).
- Relevant Coursework: Object-Oriented Programming (OOP), Data Structures & Algorithms, Software Engineering Fundamentals, Artificial Intelligence